Aseem Saxena

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SKILLS

Programming: Python (10+ years exp.), MATLAB (9+ years exp.), C/C++ (9+ years exp.), Java (9+ years exp.)
Software and Libraries: PyTorch, CUDA, Triton, OpenCV, NumPy, Pandas, Polars, JAX, SciPy, Numba, Docker, AWS, SQL

EXPERIENCE

CodeFlash, San Francisco, *Founding Machine Learning Engineer* Feb '25 – Present

- **7x Faster Inference** for Roboflow Instance Segmentation via Pipelining and Triton Kernels**[Link]**. *Triton, CUDA*
- **Faster Document Parsing** via Codeflash CLI tool**[1] [2] [3]**. *Numba, Pytorch, Jax*
- **Faster RL Environments** via Codeflash CLI tool**[1] [2] [3]**. *Numba, Pytorch, Jax*
- **30% Lower Token Costs** via Locality: Filesystem for AI agents**[1]**

Oregon State University *Graduate Research Assistant, Prof. Alan Fern* Jun '21 – Dec '24

- **Uncertainty Quantification in Vision Language Models -** Applied Conformal Prediction for calibrated evaluation of VLM model uncertainty for VQA**[Link]**.*Vision Language Models VLM, Retrieval Augmented Generation (RAG)*
- **Sim2Real RL for Bipedal Robots** - Developed an RL controller for generating gaits to reach goal foot locations.
  - Transferred to the real world from simulation via randomizing the parameters of the simulation.
  - Trained a model to accurately check if a footstep is feasible. Published at **[ICRA 2022]**. *PyTorch, Mujoco*
- **Multi-Task Learning -** Developed a Grape Cold-Hardiness probabilistic forecast model that outperforms the state-of-the-art model with just thirty seasons of data per cultivar.
  - Our work is deployed on **[AgWeatherNet]** which is used monthly by 14K subscribers.
  - **[ML Journal]**, **[AIAFS 2023]**(Accepted), **[IAAI 2023]**(Accepted). *PyTorch, RNNs*
- **AI Safety -** Proposed a Formal Criterion for avoiding Side Effects, demonstrated its effectiveness on gridworlds.
  - Published at **[NeurIPS ML Safety Workshop 2022]**. *PyTorch, AI Safety Gridworlds*
- **Offline RL -** Studied effect of different farmer strategies across different farms (without access to a simulator).
  - Trained a Multi-Dynamics World Model and showed that it incurs negative interference under limited data, undermining generalization. *World Models, Crop Simulators, Model-based Off-Policy Evaluation*
- **Teaching** - Systems Dynamics and Control, Fall 2021 with weekly office hours and evaluation duties.

Panasonic Singapore *AI Engineer, Technology Innovation Team* Jan '19 – Jan '21

- **Bayesian Optimization** for Material Design - Reduced number of iterations from 20 (2 years) to 1 (2 weeks) to obtain a material composition which meets design criteria with just 30 samples. *PyTorch, Gaussian Processes*
- **Edge Deployment** of Deep Learning Models - Successfully deployed vision models on dated Android TV boxes with lower computational resources, achieving a 30 FPS. *PyTorch, OpenCV, TensorFlow, Android 6.0, ONNX*
- Real-time **Multi-Object Tracking** - Developed a 50+ FPS tracker using Kalman Filters for state estimation and Hungarian algorithm for data association. Tracker deployed in a shop with 1000 daily visits. *OpenCV, C++*
- Deep Learning for **Gaze Estimation** - Trained a robust gaze prediction model entirely on synthetic images, fine-tuned on real images and successfully deployed on a beta trial in a shop with 1000 daily visits. *Unity, PyTorch*

National University of Singapore *Research Staff, Prof. David Hsu* Jan '17 – Jun '18

- Autonomous Driving in a Crowd by **Learning from Tree Search** - Published at **[RSS 2019]**. *PyTorch, C++, Unity*
- Developed a feature rich **Visualization Tool** to debug QMDPNet, an approx. POMDP Solver. *TensorFlow, Tkinter*
- Developed a **Robust Position and Velocity Controller** for the Fetch Robot for indoor navigation. *ROS, C++*

Ducere Technologies, India *Computer Vision Engineer* Jul '16 – Apr '17

- Developed a **Low Cost 3D LiDAR** system using TeraRanger One ToF sensor on a pan-tilt unit. *Point Cloud Library*

IIIT Hyderabad, India *Research Staff, Prof. Madhava Krishna* Apr '17- Jul '17, Jun '15 – Jul '16

- Learning based approach for **Visual Servoing -** Published at **[ICRA 2017]**. *Caffe, OpenRAVE, MATLAB, Drones*
- **Traffic Sign Detection, Recognition and Tracking** Mahindra Driverless Car Challenge. *OpenCV, C++*

COURSE PROJECTS

- Avoiding Side Effects in Conway's Game of Life Environments via Multi-Task Learning **[Slides]**
- Distributed DQN Q-Learning with Ray Framework via CPU parallelism for data collection and updates **[Code]**
- Offline RL for Bipedal Robots via Behavior Cloning and Actor-Critic Learning **[Report]**
- Studying Robustness of Semi-supervised Visual Features to Adversarial Attacks **[Report]**
- Monte Carlo Dropout for Efficient RL Exploration in Continuous Maze Environments **[Report]**

EDUCATION

Oregon State University | M.S in Artificial Intelligence | GPA: 3.89/4.0 Corvallis, OR Mar '21 - Dec '24
Courses: Reinforcement Learning, Deep Learning, Algorithms, Optimization, Probabilistic Graphical Models
Research: Multi-Task Learning, Bipedal Robots, AI Safety, Forecasting, Offline RL

Birla Institute of Technology and Science, Pilani India
B.E in Electrical & Electronics Engineering, M.Sc in Biological Sciences (Dual Major) '11- '16